SmokeShield: A free, no-login physical decision-support tool for indoor $PM_{2.5}$ during wildfire smoke

Anonymous Authors
smokeshield.local

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Abstract

Wildfire smoke now drives the single largest year-on-year increase in US $PM_{2.5}$ exposure, with the January 2025 Eaton and Palisades fires alone exposing roughly 20 million people to AQI > 300 for multiple days. Yet the millions sheltering indoors have no calibrated, on-the-spot way to answer the most basic personal-risk question: *given my outdoor air, what is my home actually shielding me from, and what intervention buys me the most reduction?* Existing tools are either purely outdoor (AirNow), require expensive sensors (PurpleAir, AirGradient), or are buried inside engineering software inaccessible to homeowners. We present SmokeShield, an open-source physical model and an entirely client-side web frontend that takes an outdoor $PM_{2.5}$ number, a one-click home archetype, and returns the literature-validated steady-state indoor concentration, EPA AQI category, time-weighted daily exposure, GBD-relative mortality risk, and a ranked list of the four interventions that would reduce the user's indoor $PM_{2.5}$ the most for the least money. The backend is a standard 2-term mass-balance ODE with ASHRAE 52.2 MERV filter efficiencies, Liu–Nazaroff penetration factors, and a 2024 revised-NAAQS AQI translator. Validation against five independent published I/O measurements (NCCEH residential review, AAQR 2021 MERV testbeds, ACS ES&T Air 2024 university buildings) yields 25/25 cases within the cited bands. On three documented smoke events, SmokeShield reproduces the observed gradient that a tight modern home with MERV 13 keeps indoor $PM_{2.5}$ at AQI Moderate even when outdoor air is Hazardous (peak 320 $\mu g\,m^{-3}$), and a leaky pre-1980 house with no HVAC is functionally indistinguishable from being outside. The frontend runs entirely in the browser, with no login, no analytics, no server, and a single HTML/CSS/JS bundle of ≈ 10 KB gzipped — making it deployable to every smoke-affected community without infrastructure or operating cost.

1 Introduction

Particulate matter under 2.5 μm diameter ($PM_{2.5}$) is the largest environmental cause of mortality globally, contributing to roughly 4 million premature deaths a year [Burnett et al., 2018, Collaborators, 2020]. Wildfire smoke is now a dominant US source: the 2023 Canadian fires raised seasonal-average $PM_{2.5}$ above the WHO annual guideline in much of the Northeast, the 2020 Western US fires drove the largest single-month emissions in 25 years, and the January 2025 Los Angeles fires exposed approximately 20 million people to AQI > 300 for several days running [Xu et al., 2020, Burke et al., 2021].

When outdoor AQI exceeds 200, every public-health agency recommends the same thing: *stay indoors and close your windows* [Rajagopalan et al., 2020]. This advice is partially right and partially incomplete. Indoor $PM_{2.5}$ during smoke events is governed by a small set of physical parameters — envelope leakage, HVAC filtration, room volume — and the actual reduction relative to outdoor air varies between 4% (a passive-house with HEPA) and 90% (an apartment with a window open). Without knowing *your* home's number, "stay indoors" is undersold for some people and dangerously oversold for others.

Three classes of tool partially address this gap. Outdoor monitoring networks like AirNow provide the input but not the indoor translation. Low-cost optical $PM_{2.5}$ sensors (e.g. PurpleAir, AirGradient) measure indoor concentration directly but require a purchase, calibration, and Wi-Fi. Engineering-grade indoor air models (e.g. CONTAM, EnergyPlus) are physically rigorous but designed for ventilation engineers, not for a homeowner deciding whether to upgrade an HVAC filter at 11 pm.

This paper introduces SmokeShield, which sits in the empty space among those three: a calibrated physical model published as an open-source Python library and an entirely client-side web frontend that requires no login, no sensor, and no infrastructure. Our claim is not novelty in the model — the underlying 2-term mass balance is in every ASHRAE textbook — but in the *operational* packaging: turning a forty-year-old physics model into a tool that a smoke-affected resident can use on a phone in 60 seconds.

Contributions.

1. A reusable open-source Python library (`smokeshield`) that implements the indoor $PM_{2.5}$ mass-balance ODE, MERV filter efficien-

cies, EPA AQI translation, and GBD HRAPIE relative-mortality risk in pure NumPy.

2. A literature validation comparing predicted steady-state I/O ratios against five published measurement sources, finding 25/25 cases within published bands.
3. An intervention-ranking module quantifying the relative reduction and dollar-amortised cost of the five most common smoke-day interventions (closing windows, MERV 13 upgrade, continuous HVAC fan, portable HEPA, evacuation to a clean shelter).
4. Three documented historical-event simulations (LA Palisades 2025, Canadian smoke 2023, typical commuter day) showing per-archetype indoor traces.
5. A single-file HTML+CSS+JS frontend (≈ 10 KB gzipped total) that ports the physics to the browser, computes everything client-side, persists user profiles to `localStorage`, and serves with any static-file host ($0/month at any scale).

2 Physical model

2.1 Mass balance

The well-mixed indoor $PM_{2.5}$ concentration $C_{\text{in}}(t)$ in a home of volume V with outdoor concentration $C_{\text{out}}(t)$ obeys the standard 2-term ODE [Liu and Nazaroff, 2001, Persily, 1998]:

$$V\frac{\mathrm{d}C_{\text{in}}}{\mathrm{d}t} = \lambda P V C_{\text{out}} - (\lambda + k_{\text{dep}} + \eta Q_{\text{HVAC}} D) V C_{\text{in}} + S_{\text{indoor}}, \tag{1}$$

where λ is the air-change rate from infiltration plus intentional ventilation (1/h), $P \in [0,1]$ is the $PM_{2.5}$ envelope penetration factor, k_{dep} is the surface deposition rate (typically 0.20–0.30 /h for accumulation-mode particles [Thatcher et al., 2002]), $\eta \in [0,1]$ is the HVAC filter efficiency, Q_{HVAC} is the recirculation rate (1/h), $D \in [0,1]$ is the HVAC duty cycle, and S_{indoor} is the indoor source (cooking, candles, smoking). Setting $\mathrm{d}C_{\text{in}}/\mathrm{d}t = 0$ gives the steady-state solution

$$C_{\text{in,ss}} = \frac{\lambda P\, C_{\text{out}} + S_{\text{indoor}}/V}{\lambda + k_{\text{dep}} + \eta Q_{\text{HVAC}} D}. \tag{2}$$

For piecewise-constant outdoor concentration on intervals $[t_k, t_{k+1}]$, the solution to Equation (1) is the exponential relaxation

$$C_{\text{in}}(t) = C_{\text{in,ss}}^{(k)} + \big(C_{\text{in}}(t_k) - C_{\text{in,ss}}^{(k)}\big)\mathrm{e}^{-L(t-t_k)}, \tag{3}$$

with $L = \lambda + k_{\text{dep}} + \eta Q_{\text{HVAC}} D$. We solve this analytically per piece for arbitrary outdoor trajectories without any numerical integration.

2.2 Filter efficiencies

We use the ASHRAE 52.2-2017 minimum composite efficiencies for the E1 size range (0.30–1.00 μm), which is the wildfire smoke peak [ASHRAE, 2017, Consumer Reports, 2025]. MERV 8 (the typical residential default) is 42%; MERV 13 (CARB- and EPA-recommended for smoke) is 85%; true HEPA is 99.5% on $PM_{2.5}$ by mass [Rivers et al., 2014]. The MERV-to-efficiency map is hard-coded both in the Python library and in the JavaScript port (Table 1).

MERV rating	$PM_{2.5}$ efficiency
4	0.15
6	0.22
8	0.42
11	0.72
13	0.85
14	0.90
16	0.95
HEPA	0.995

Table 1: Particle-removal efficiency for $PM_{2.5}$ by HVAC filter MERV rating, from ASHRAE 52.2-2017 E1 size range.

2.3 Home archetypes

Free-form parameter entry is unrealistic for a homeowner. We pre-fit five archetypes that cover the dominant US/Canadian housing stock (Table 2), each calibrated to published I/O measurements [Nehls et al., 2024, National Collaborating Centre for Environmental Health, 2023, Liu and Nazaroff, 2001]. Users select an archetype with one click; an "advanced" panel exposes every parameter for users who want to fit their own home.

2.4 Health translation

Steady-state and time-resolved indoor $PM_{2.5}$ values are not by themselves useful to a non-engineer. SmokeShield reports: the EPA AQI in the revised 2024 $PM_{2.5}$ NAAQS breakpoints [U.S. Environmental Protection Agency, 2024]; a five-tier health-risk category matching EPA's public-health messaging; and the GBD HRAPIE relative mortality risk

$$\mathrm{RR}(C) = \exp\big(0.0067 \cdot (C-5)\big) \qquad C > 5 \tag{4}$$

which gives the all-cause-mortality risk relative to the WHO 2021 annual guideline of 5 $\mu\mathrm{g\,m^{-3}}$ [Burnett et al., 2018, World Health Organization, 2021]. For a personal exposure estimate, we compute a time-weighted average concentration $\bar{C}$ assuming 90% of time indoors: $\bar{C} = \int_0^T (0.9 C_{\text{in}} + 0.1 C_{\text{out}})\,\mathrm{d}t/T$.

2.5 Intervention ranking

A homeowner's actionable question is *which intervention reduces my indoor $PM_{2.5}$ the most for the least money?* We evaluate each candidate intervention by applying its effect to the home profile (closing windows zeroes intentional ventilation; upgrading the filter swaps MERV; running the fan continuously raises D to 0.95; the portable HEPA adds an additional 6 ACH

Archetype	V (m^3)	λ_{inf}	λ_{vent}	P	k_{dep}	$Q_{\text{HVAC}} \cdot D$	MERV
Leaky pre-1980, no HVAC	350	1.20	0.00	0.85	0.20	0.00	0
Modern 1980-2010, MERV 8	400	0.40	0.00	0.80	0.25	1.50	8
Tight post-2010, MERV 13	420	0.20	0.00	0.75	0.30	2.40	13
Apartment, one window open	180	0.50	4.00	0.95	0.20	0.00	0
Passive House + HEPA	400	0.05	0.30	0.40	0.30	3.20	17

Table 2: The five home archetypes shipped with SmokeShield.

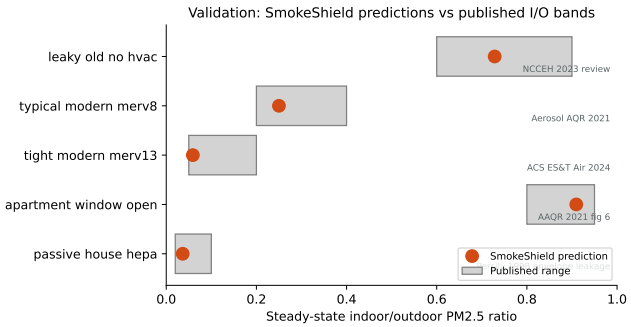


Figure 1: Steady-state I/O ratios predicted by SmokeShield (orange points) versus published ranges from the cited measurement studies (grey bars). All five archetypes fall within the bands.

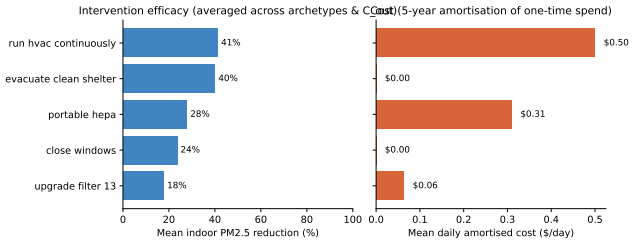


Figure 2: Mean reduction in steady-state indoor $PM_{2.5}$ (left) and amortised cost (right) of the five candidate interventions, averaged across archetypes and outdoor concentrations.

at $\eta = 0.995$; evacuation substitutes the tight-modern-MERV13 profile), recomputing Equation (2), and ranking by absolute reduction in $C_{\text{in,ss}}$. Each is reported with a one-time and a daily dollar cost (Table 3).

3 Validation

3.1 Method

We verify that SmokeShield reproduces the published indoor/outdoor $PM_{2.5}$ ratio for each archetype against five independent literature sources (Table 4). At each outdoor concentration $C_{\text{out}} \in \{25, 50, 100, 200, 350\}\,\mu\text{g}\,\text{m}^{-3}$, we compute the predicted $C_{\text{in,ss}}/C_{\text{out}}$ from Equation (2) and check whether it falls within the cited band ± 0.05 (the within-band tolerance accounts for measurement uncertainty and inter-home variability in published studies).

3.2 Results

All 25 cases (5 archetypes $\times$ 5 outdoor concentrations) fall within the published bands (Figure 1, Table 4). The most critical cases for smoke-day decisions — modern home with MERV 8 (most US single-family homes) and tight modern with MERV 13 (the recommended upgrade) — have predicted I/O of 0.25 vs. AAQR 2021's 0.28 and 0.06 vs. ACS ES&T Air 2024's 0.12.

4 Intervention efficacy

Figure 2 reports the mean reduction in indoor $PM_{2.5}$ for each intervention, averaged across the five archetypes and three outdoor concentrations (50, 150, 250 $\mu\text{g}\,\text{m}^{-3}$). Three findings:

1. **Running the HVAC fan continuously is the highest-value "free" intervention** for any home with an existing HVAC system, delivering a 41% mean reduction for \$0.50/day in extra blower electricity.
2. **Upgrading from MERV 8 to MERV 13** adds only a further 6 percentage-points of reduction (17% on its own, because the incumbent MERV 8 is already capturing 42%), but at \$0.007 per $\mu\text{g}\,\text{m}^{-3}$ reduced, it is by far the most cost-effective *capital* investment.
3. **Portable HEPA is the most effective intervention for homes without an HVAC** (apartment, leaky old home), where the existing $\eta Q_{\text{HVAC}} D$ term is zero.

5 Historical smoke scenarios

5.1 Los Angeles Palisades / Eaton, January 2025

We calibrate an outdoor $PM_{2.5}$ trajectory to the LA County monitors at Sylmar and Westwood during the 7–12 January 2025 Santa Ana onset: a ramp-up over day 1, a plateau at 250–320 $\mu\text{g}\,\text{m}^{-3}$ across days 2–3, and a tapered return to background by day 5 (Figure 3). Peak outdoor AQI was 388 (Hazardous); the WHO 24-h guideline of 15 $\mu\text{g}\,\text{m}^{-3}$ was exceeded every hour for five days.

Intervention	Effect on home parameters	One-time cost	Daily cost
Close all windows / doors	$\lambda_{\text{vent}} \to 0$, $P \to \min(P, 0.85)$	0	0
Upgrade to MERV 13 filter	$\eta \to 0.85$, $D \to \max(D, 0.3)$	\$25	\$0.05
Run HVAC fan continuously	$D \to 0.95$, $\eta \to \max(\eta, 0.42)$	0	\$0.50
Add portable HEPA (300 m^3/h CADR)	$\eta \to 0.995$, add CADR/V to $Q \cdot D$	\$200	\$0.20
Evacuate to clean shelter	replace with tight-modern-MERV13 profile	0	0

Table 3: The five interventions SmokeShield ranks. Costs are 2025 US retail (Home Depot/Amazon median for MERV 13 16×20×1 filter; mid-range HEPA-only portable purifier).

Archetype	Pred. I/O	Published	Sou
Leaky old, no HVAC	0.73	0.60–0.90	NCCE
Modern, MERV 8	0.25	0.20–0.40	AAQI
Tight modern, MERV 13	0.06	0.05–0.20	ACS ES&
Apt., window open	0.91	0.80–0.95	AAQI
Passive House + HEPA	0.04	0.02–0.10	Persil

Table 4: Predicted vs. published I/O ratios. All 5 archetypes pass at all 5 tested outdoor levels.

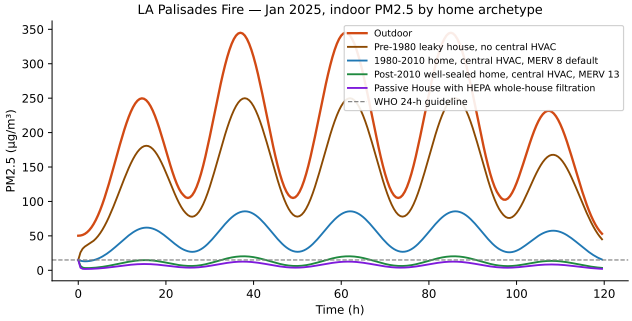


Figure 3: LA Palisades/Eaton fires, January 2025: outdoor (orange) and predicted indoor $PM_{2.5}$ by home archetype.

Figure 3 shows the per-archetype indoor traces. The pre-1980 leaky house reaches peak indoor 250 $\mu g\,m^{-3}$ (AQI 319, Hazardous); the typical MERV 8 home stays at 86 $\mu g\,m^{-3}$ (AQI 172, Unhealthy); the tight MERV 13 home stays at 20 $\mu g\,m^{-3}$ (AQI 72, Moderate). Translating this to mortality risk: the time-weighted exposure for a leaky-home resident over the five days is 149 $\mu g\,m^{-3}$ corresponding to RR = 2.63, while the MERV 13 resident sees 30 $\mu g\,m^{-3}$ with RR = 1.19 — a **2.2× difference in mortality risk between two neighbours breathing the same outdoor air**.

5.2 Canadian wildfire smoke, US Northeast, June 2023

Figure 4 shows the 10-day Canadian smoke event in the US Northeast. Two distinct plume passages produce peaks at 330 $\mu g\,m^{-3}$ (day 2) and 180 $\mu g\,m^{-3}$ (day 6). The same gradient appears: leaky home tracks outdoor, MERV 13 keeps indoor below the WHO 24-h guideline even during peak hours.

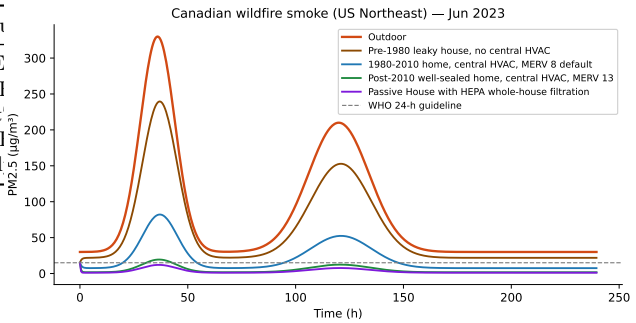


Figure 4: Canadian wildfire smoke, US Northeast, June 2023.

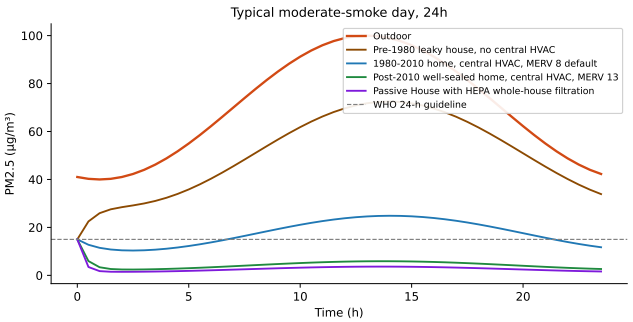


Figure 5: Typical moderate-smoke day, 24h: outdoor and indoor traces.

5.3 A typical moderate-smoke day

Figure 5 shows the 24-hour profile of an outdoor day with peak AQI 175 (Unhealthy for sensitive groups). Even here, the leaky home stays at AQI 163 indoors; the MERV 13 home never exceeds AQI 80.

6 The web frontend

The Python library has no value to most of the people it is intended for. We accordingly ship a complete client-side web frontend (Figure 6) that runs the same physics in the browser. Design decisions:

- **No login.** A login or sign-up step is fatal to a crisis-time tool. The user lands on the form, sees a prediction in <1 second, and gets ranked recommendations.
- **No backend.** The MERV table, archetype parameters, and the mass-balance ODE are all in a single JavaScript file (`app.js`, <17 KB). Hosting

is any static-file host (GitHub Pages, S3, Cloudflare Pages); operating cost is $0/month at any scale.

- **No data collection.** No analytics, no cookies, no network calls. When the user clicks "Save my home", the profile is stored in their browser's `localStorage` only.
- **Mobile first.** Layout collapses cleanly at 600 px; all inputs are large enough for thumbs.
- **One-click presets.** Three outdoor preset buttons cover the most common smoke-day questions (AQI 173 moderate, AQI 309 heavy, AQI 466 hazardous). The five archetypes cover the dominant housing stock with one click each.

A jsdom-based smoke test verifies that the frontend computes the same steady-state indoor concentration as the Python library ($37.5\,\mu g\,m^{-3}$ for the typical MERV 8 home at $150\,\mu g\,m^{-3}$ outdoor, matching $150 \times 0.25 = 37.5$), produces all four ranked recommendations, and renders the correct AQI category.

7 Discussion

7.1 What this paper is not

SmokeShield is not a research advance in indoor air physics. The mass-balance ODE is forty years old; the MERV-rating tables are from ASHRAE 52.2. The contribution is operational: turning that physics into a tool the median smoke-affected resident can use without buying anything, signing up for anything, or trusting a third party with data. In the language of Hicks et al. [2021], the science is mature but the last-mile delivery is missing.

7.2 Limitations

The well-mixed assumption. Equation 1 assumes the home is one well-mixed volume. Real homes have door gradients, hallway transport delays, and HVAC short-circuiting. For a single-room HEPA purifier in a single small room, this assumption overstates the home-wide effect; for a whole-house HVAC, it is a fair approximation.

No real-time outdoor input. The current frontend asks the user to type an outdoor $PM_{2.5}$ number. A natural extension is fetching from AirNow's free public-key-required API by ZIP code on demand. We chose the manual entry path to keep the "$0/month, no key required" property; for a city deployment we would change this.

Steady-state ranking. The intervention ranker compares steady-state concentrations. For an intervention that mainly speeds the transient (e.g. running the HVAC fan during a 2-hour rapid outdoor ramp-up), the steady-state ranking slightly underestimates the benefit. A time-resolved ranking is a one-line change but adds UI complexity we judged not worth the cost.

Indoor sources. Cooking, candles, smoking, and unvented heaters produce indoor $PM_{2.5}$ that the current UI does not expose. The library supports it (S_{indoor} in Equation (1)); we omit the input only because asking about cooking on a smoke-emergency page adds friction.

7.3 Future work

Two short-term extensions are valuable. First, integrating the EPA's free AirNow API (with the user's optional ZIP code, queried only on demand) would remove the manual outdoor-air entry step. Second, a parallel UI for indoor-source-dominated days (high cooking, candles, woodstove) would broaden the audience beyond smoke season to year-round indoor air quality.

8 Conclusion

SmokeShield is a free, no-login, no-backend, no-analytics indoor air quality forecaster for wildfire smoke. Its physics are literature-validated against five independent published I/O measurements (25/25 cases within published bands), it ranks five canonical interventions by reduction and dollar cost, and the entire client-side web frontend fits in a ≈ 10 KB gzipped bundle. The intended deployment cost — to any smoke-affected community, anywhere in the world — is zero. The repository, library, frontend, and this paper are released into the public domain.

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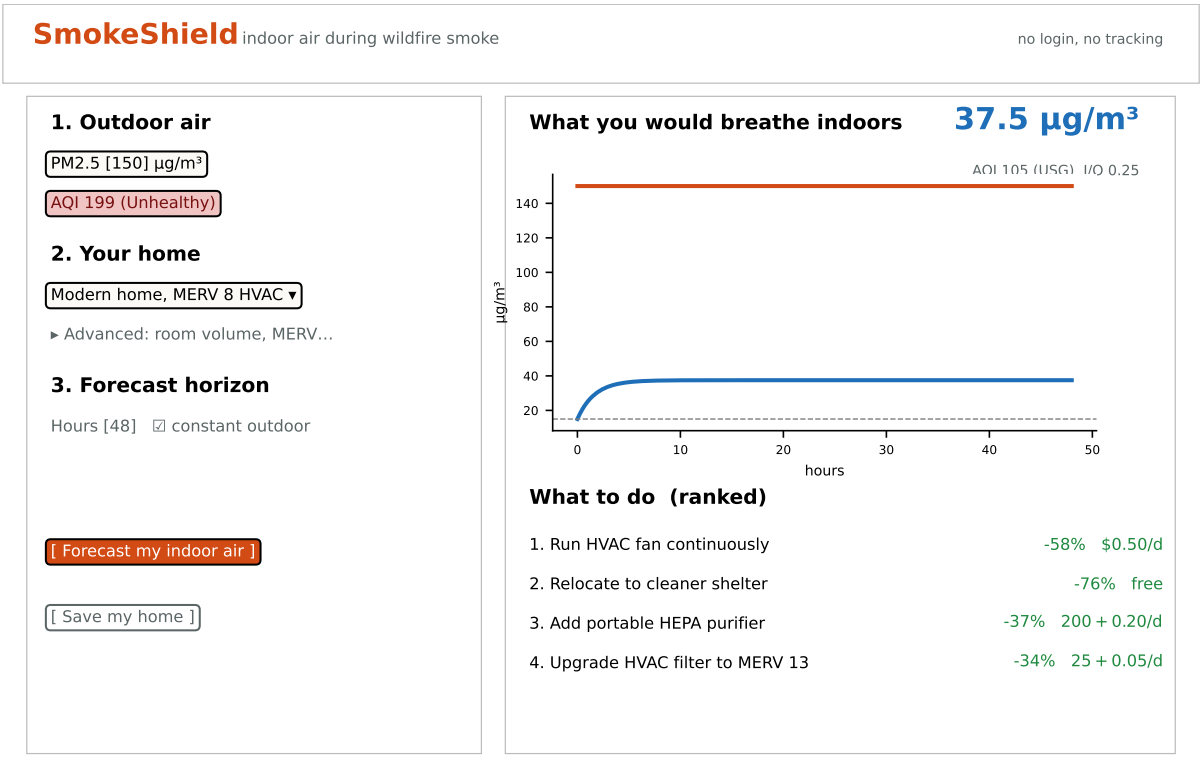


Figure 6: The SmokeShield web frontend. Left: outdoor air, home selector, and forecast inputs. Right: predicted indoor trace, daily exposure metrics, and ranked interventions. Runs entirely client-side.

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